

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Final Exam  
 Duration: 1 Hour 30 Minutes

Semester : Summer 2022  
 Full Marks: 30

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1. Dhaka, especially your area, is facing an Electricity crisis. Many initiatives including load shedding, reducing office hours, etc have already been taken. The local authority of your area is looking for feasible solutions to reduce the consumption of electricity. One of the members of their advisory committee has suggested that they should reduce the operating costs of road lighting. Till now every road is illuminated all night long. To reduce electricity consumption, they have decided to no longer illuminate every road, but to switch off the road lighting of some roads. To make sure that the inhabitants of your area still feel safe, they want to optimize the lighting in such a way that after darkening some roads at night, there will still be at least one illuminated path from every major point in your area to every other major point.

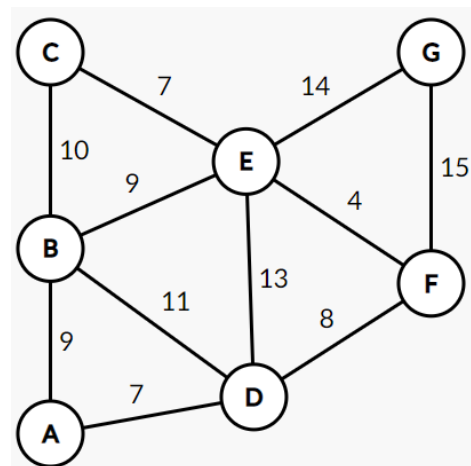


Figure : Graph of your area where the Vertices denote major points and Edges denote electricity consumption in a day.

- a. **Apply** a suitable Algorithm to help the authority determine the roads they need to illuminate and the cost which will minimize the electricity consumption. If you need to consider any root vertex for your Algorithm, you can consider 'A' as such. **6**
- CO3
- b. Observe the Graph of your area and your provided solution of question 1(a) and **determine** the maximum electricity of a day the authority can save. **1**
- CO5
- c. The United Arab Emirates (UAE), especially Dubai, is hosting the Asia Cup 2022. They too want to minimize the electricity consumption by switching off the lights in some roads leaving exactly one illuminated path from every major point in your area to every other major point. However, to keep the tourists entertained, the Authority of Dubai has taken an opposite approach to yours. They want to illuminate the roads which require the most electricity. **3**
- CO6
- Propose** an algorithm to compute the path which fulfills this requirement.

2. Alice and Bob both are the peers of a decentralized distributed network. Alice wants to send a message to Bob using the platform but she does not want that any other users of the network would be able to understand the message. That is why she encodes the message using an encoding algorithm which is known to only both of them and sends it to Bob. The message is:

**Abaaccdd efkkslasae,eEklS aaad.**

- a. **Simulate** the Huffman Encoding algorithm to encode the message. Construct Huffman Tree, generate the codeword for each character, show the encoded message and count the total number of bits required to store the message. **7**  
CO4
- b. **Explain** the advantage of Huffman Encoding over the ascii codes approach mathematically. **1**  
CO5
- c. After reading the message Bob replied to Alice by sending an encoded message: **2**  
CO4 10001010101111101010.  
Now **use** the Huffman Tree constructed in question 2(a) to decode the message.

3. Today Alice has learned how to find the Longest Common Subsequence (LCS) of two given strings. Now she wants to find the LCS of “axyb” and “abyxb”. After hours of hard work, she has made this LCS table, M.

|       | empty | a | b | y | x | b |
|-------|-------|---|---|---|---|---|
| empty | 0     | 0 | 0 | 0 | 0 | 0 |
| a     | 0     | 1 | 1 | 1 | 1 | 1 |
| x     | 0     | 1 | 1 | 1 | 2 | 2 |
| y     | 0     | 1 | 1 | 2 | 2 | 2 |
| b     | 0     | 1 | 2 | 2 | 2 | 3 |

- a. **Explain** what you understand by the value  $M[3][4] = 2$ . **2**  
CO1
- b. **Find** out the LCS String from the table. Show the steps of your work. **3**  
CO2
- c. **Determine** the maximum profit for the 0-1 Knapsack problem given in the following table **5**  
CO4 using Dynamic Programming. Show the steps with a recursion tree or the memory matrix.

Knapsack Weight: 8 kg

| Objects     | <i>i</i> | <i>ii</i> | <i>iii</i> | <i>iv</i> |
|-------------|----------|-----------|------------|-----------|
| Weight (kg) | 5        | 4         | 6          | 3         |
| Profit (\$) | 11       | 10        | 12         | 9         |

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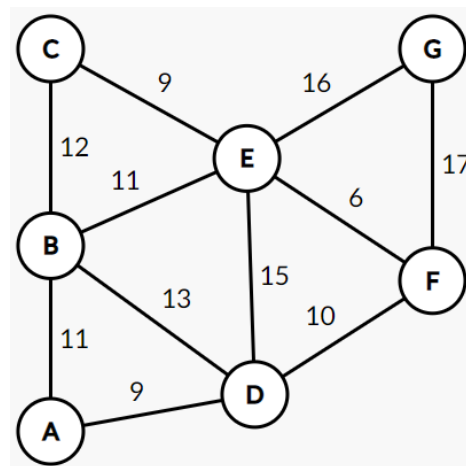


Figure : Graph of your area where the Vertices denote major points and Edges denote electricity consumption in a day.

- a. **Apply** a suitable Algorithm to help the authority determine the roads they need to illuminate and the cost which will minimize the electricity consumption. If you need to consider any root vertex for your Algorithm, you can consider 'A' as such. **6**
- CO3
- b. Observe the Graph of your area and your provided solution of question 1(a) and **determine** the maximum electricity of a day the authority can save. **1**
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- c. The United Arab Emirates (UAE), especially Dubai, is hosting the Asia Cup 2022. They too want to minimize the electricity consumption by switching off the lights in some roads leaving exactly one illuminated path from every major point in your area to every other major point. However, to keep the tourists entertained, the Authority of Dubai has taken an opposite approach to yours. They want to illuminate the roads which require the most electricity. **3**
- CO6
- Propose** an algorithm to compute the path which fulfills this requirement.

2. Alice and Bob both are the peers of a decentralized distributed network. Alice wants to send a message to Bob using the platform but she does not want that any other users of the network would be able to understand the message. That is why she encodes the message using an encoding algorithm which is known to only both of them and sends it to Bob. The message is:

**Saaees eklasddd,Aeeesss lkes.**

- a. **Simulate** the Huffman Encoding algorithm to encode the message. Construct Huffman Tree, generate the codeword for each character, show the encoded message and count the total number of bits required to store the message. 7  
CO4
- b. **Explain** the advantage of Huffman Encoding over the ascii codes approach mathematically. 1  
CO5
- c. After reading the message Bob replied to Alice by sending an encoded message: 2  
CO4 1111110000111011010.  
Now **use** the Huffman Tree constructed in question 2(a) to decode the message.

3. Today Alice has learned how to find the Longest Common Subsequence (LCS) of two given strings. Now she wants to find the LCS of “axyb” and “axyxb”. After hours of hard work, she has made this LCS table, M.

|       | empty | a | x | y | x | b |
|-------|-------|---|---|---|---|---|
| empty | 0     | 0 | 0 | 0 | 0 | 0 |
| a     | 0     | 1 | 1 | 1 | 1 | 1 |
| x     | 0     | 1 | 2 | 2 | 2 | 2 |
| y     | 0     | 1 | 2 | 3 | 3 | 3 |
| b     | 0     | 1 | 2 | 3 | 3 | 4 |

- a. **Explain** what you understand by the value  $M[3][4] = 3$ . 2  
CO1
- b. **Find** out the LCS String from the table. Show the steps of your work. 3  
CO2
- c. **Determine** the maximum profit for the 0-1 Knapsack problem given in the following table 5  
CO4 using Dynamic Programming. Show the steps with a recursion tree or the memory matrix.

Knapsack Weight: 8 kg

| Objects     | <i>i</i> | <i>ii</i> | <i>iii</i> | <i>iv</i> |
|-------------|----------|-----------|------------|-----------|
| Weight (kg) | 5        | 4         | 6          | 3         |
| Profit (\$) | 9        | 8         | 20         | 9         |

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Examination: Final Exam  
 Duration: 1 Hour 40 Minutes

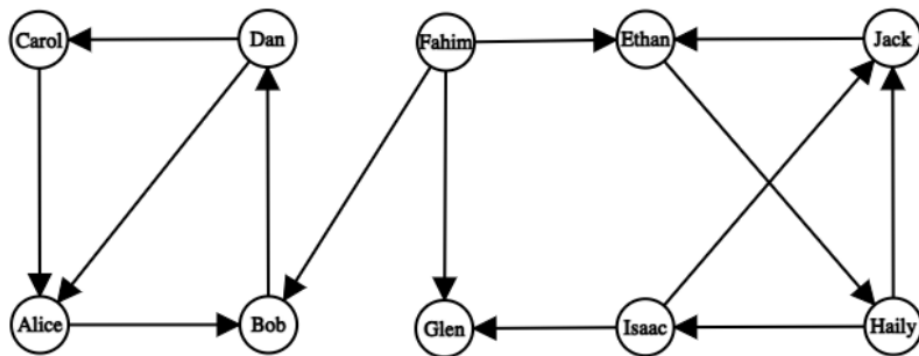
Semester: Summer 2023  
 Full Marks: 40

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1. Among a group of 10 friends, it's not feasible for each individual to visit the houses of all the others. With an upcoming tournament in mind, the friends need to establish groups. The criterion for forming these groups is that a friend must be capable of visiting the houses of every other member in their group. The task involves calculating the count of viable groups following this criterion. A group has to consist of a minimum of two people.



- |                                                                                                                                                                                                      |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| a) [CO1] The end result will showcase the list of groups and the members. <b>Name</b> the algorithm tailored for this situation.                                                                     | <b>01</b> |
| b) [CO1] <b>Show</b> a simulation of your chosen algorithm using the graph above. <b>Write</b> the number of groups that can be formed and the members of each group.                                | <b>05</b> |
| c) [CO1] Those who are incapable of forming a group will be identified as individuals ineligible for tournament participation. <b>Identify</b> those people if there are any.                        | <b>01</b> |
| d) [CO3] We also want to find the person who can visit the most houses. <b>Propose</b> an algorithm to implement this. Present your algorithm with a pseudocode/flowchart/step-by-step instructions. | <b>03</b> |

**Answer only one from 2, 2-OR**

2. Consider the following modified ASCII scheme and the encoded message using it.

| Character | a    | b    | c    | d    | e    | f    | g    | h    | i    | j    |
|-----------|------|------|------|------|------|------|------|------|------|------|
| m_ascii   | 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 |

Message encoded with m\_ascii code:

0000 0100 0100 1000 0100 1000 0100 1000 0001 0101 0001 0001 0110 0000 0100 1001 0010 1001

Now answer the following questions.

- a) [CO1] **Decode** the message. 01
- b) [CO1] **Show** simulation to find the encoding of each of the characters using Huffman coding technique. While creating a tree, less frequency nodes go to the left side of the root; left is considered as a 0, right is 1. 06
- c) [CO1] **Calculate** the number of bits required in the encoded message using Huffman technique. 03

**OR**

- 2.** You study in a university and there are some faculties in your university and their class schedule (start and end time) are given below. As a pantomath (who wants to learn a lot) person you want to gather more knowledge and do class of the maximum faculties possible. But, due to the time conflict, one student cannot cover all the classes. So, you asked your friends Derke, Leo and Chronicle how you can maximize the number of classes. Derke said you should choose the classes which have the lowest duration and Leo said you should choose those classes which end early and Chronicle said you should pick classes that start early.

- a) [CO1] Who's method will you **select** in order to complete the classes of maximum faculties? 01
- b) [CO1] Using the method you mentioned in (a) **simulate** and find out the maximum number of classes you can do from the following schedule and also find out the initials of the faculties. 06

| Faculty | Start Time | End Time |
|---------|------------|----------|
| MIBA    | 1          | 10       |
| MZU     | 2          | 3        |
| AGD     | 6          | 8        |
| MNR     | 10         | 11       |
| RIM     | 5          | 7        |
| FGZ     | 3          | 6        |
| SBD     | 7          | 10       |

- c) [CO1] As one student cannot cover all the classes so you want to determine the minimum number of students needed so that all the classes are covered. **Explain** your algorithm in a pseudocode/flow-chart/step-by-step instructions format. 03

- 3.** Karen started a website that displays different advertisements. She wants to maximize her advertisement revenue. Each month she receives a list of advertisements, each with a certain value (revenue) and a size (space taken on the webpage). The webpage has a limited amount of space, and she selects a subset of advertisements to display in order to maximize her total revenue, while staying within the available space. No advertisement is selected more than once, and each advertisement that is selected has to be displayed in full.

For example, following is a list of advertisements she received for the month of August:

|                                                                                                                                                                         |                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Advertisements:<br>Ad 1 - Value: \$10, Size: 2<br>Ad 2 - Value: \$8, Size: 1<br>Ad 3 - Value: \$15, Size: 3<br>Ad 4 - Value: \$6, Size: 2<br>Maximum Space Available: 5 | Solution:<br>Select Advertisements: Ad 1, Ad 3<br>Total Revenue: \$10 + \$15 = \$25<br>Space Occupied = 2 + 3 = 5 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|

- a) [CO1] Suppose, you are trying to come up with a Dynamic Programming approach to solve this problem. You build a table, each cell representing the solution (optimal revenue value) to a subproblem. **Give** the formula for filling up each entry of the table. **02**
- b) [CO1] Suppose in the month of September, she received the following list of advertisements: **06**

Advertisements:

Ad 1 - Value: \$12, Size: 4

Ad 2 - Value: \$8, Size: 2

Ad 3 - Value: \$9, Size: 3

Ad 4 - Value: \$7, Size: 2

Ad 5 - Value: \$5, Size: 1

The maximum available space is 5.

**Fill** the following table according to the formula you developed in question (a). Then write the set of advertisements Karen should select.

| Value | Size | Item | Maximum Available Space |   |   |   |    |    |
|-------|------|------|-------------------------|---|---|---|----|----|
|       |      |      | 0                       | 1 | 2 | 3 | 4  | 5  |
| -     | -    | 0    |                         |   |   |   |    |    |
| 12    | 4    | 1    |                         |   |   |   |    |    |
| 8     | 2    | 2    | 0                       | 0 | 8 | 8 | 12 | 12 |
| 9     | 3    | 3    |                         |   |   |   |    |    |
| 7     | 2    | 4    |                         |   |   |   |    |    |
| 5     | 1    | 5    |                         |   |   |   |    |    |

- c) [CO3] Suppose, Karen discovered that the top half of some of the advertisements contain nothing. So, from the month of October she is implementing a new feature, the size of at most K items can be changed to half of its original size. Revenue will stay the same. For example, **02**

There are four advertisements to select from,

Value = [10, 12, 8, 6]

Required Space = [4, 5, 6, 6]

Max Available Space = 7

K = 1

**Output: 22**

Explanation: Without changing the required space of any item, the solution would be taking only Ad-2, giving a revenue of 12. But if we change the size of Ad-1 into half of its original size, we can take both Ad-1 and 2. Then the total revenue is 22, which is the maximum.

Think about how you would solve this problem using a Dynamic Programming approach. Then, **write** a recursive formulation of your solution or **explain** your idea in pseudocode/flow-chart/step-by-step instructions format.

4. The white council once decided to install bidirectional rail tracks to connect several realms of middle earth. Upon their request, the elves and the dwarves devised a plan. The following table lists the tracks that can be constructed.

| From     | To       | Length of the track | Construction cost | Construction time |
|----------|----------|---------------------|-------------------|-------------------|
| Arnor    | Fangorn  | 25                  | 70                | 80                |
| Arnor    | Lindon   | 20                  | 40                | 30                |
| Arnor    | Mirkwood | 30                  | 80                | 70                |
| Fangorn  | Gondor   | 10                  | 30                | 40                |
| Fangorn  | Lindon   | 40                  | 90                | 110               |
| Fangorn  | Mirkwood | 20                  | 60                | 50                |
| Gondor   | Harad    | 25                  | 100               | 90                |
| Gondor   | Lindon   | 45                  | 110               | 100               |
| Gondor   | Mirkwood | 30                  | 50                | 60                |
| Gondor   | Rhun     | 40                  | 120               | 130               |
| Harad    | Mirkwood | 45                  | 150               | 180               |
| Mirkwood | Rhun     | 25                  | 180               | 150               |

As there is only one engine available for the task, simultaneous construction of multiple tracks is not possible. Going through the plan, the council decided that the following tracks **must be constructed**.

- Gondor to Lindon
- Arnor to Mirkwood

However, due to a financial crisis, the council wants to connect all the realms with **minimum total construction cost**. Now answer the following questions.

- a) [CO1] Which other tracks should they select for construction? **Show** a simulation of your procedure. **05**
- b) [CO1] **Write** the total construction cost of the selected tracks. **01**
- c) [CO1] **Calculate** the total construction time required if no two tracks are constructed simultaneously. **01**
- d) [CO1] After these tracks have been constructed, what would be the minimum distance of Rhun from all other realms? Just **write** the distance from each realm. **03**



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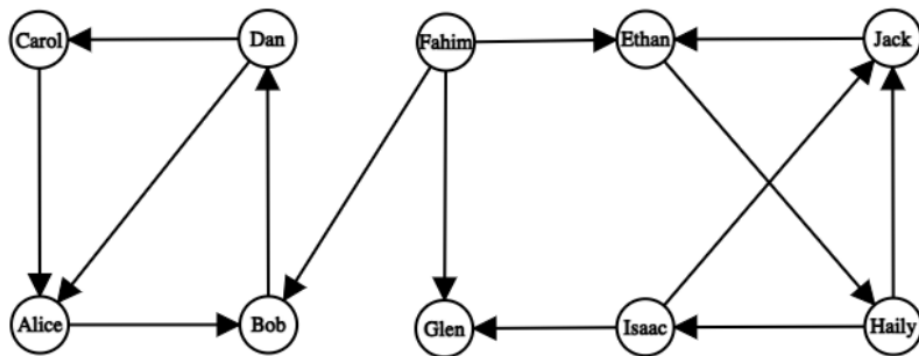
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1. Among a group of 10 friends, it's not feasible for each individual to visit the houses of all the others. With an upcoming tournament in mind, the friends need to establish groups. The criterion for forming these groups is that a friend must be capable of visiting the houses of every other member in their group. The task involves calculating the count of viable groups following this criterion. A group has to consist of a minimum of two people.



- |                                                                                                                                                                                                      |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| a) [CO1] The end result will showcase the list of groups and the members. <b>Name</b> the algorithm tailored for this situation.                                                                     | <b>01</b> |
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| c) [CO1] Those who are incapable of forming a group will be identified as individuals ineligible for tournament participation. <b>Identify</b> those people if there are any.                        | <b>01</b> |
| d) [CO3] We also want to find the person who can visit the most houses. <b>Propose</b> an algorithm to implement this. Present your algorithm with a pseudocode/flowchart/step-by-step instructions. | <b>03</b> |

**Answer only one from 2, 2-OR**

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|-----------|------|------|------|------|------|------|------|------|------|------|
| m_ascii   | 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 |

Message encoded with m\_ascii code:

0010 0000 0010 0000 0011 0110 0111 1000 0011 1001 0000 0010 0011 0010 1001 1000 0010 0111

Now answer the following questions.

- a) [CO1] **Decode** the message. 01
- b) [CO1] **Show** simulation to find the encoding of each of the characters using Huffman coding technique. While creating a tree, less frequency nodes go to the left side of the root; left is considered as a 0, right is 1. 06
- c) [CO1] **Calculate** the number of bits required in the encoded message using Huffman technique. 03

OR

2. You study in a university and there are some faculties in your university and their class schedule (start and end time) are given below. As a pantomath (who wants to learn a lot) person you want to gather more knowledge and do class of the maximum faculties possible. But, due to the time conflict, one student cannot cover all the classes. So, you asked your friends Derke, Leo and Chronicle how you can maximize the number of classes. Derke said you should choose the classes which have the lowest duration and Leo said you should choose those classes which end early and Chronicle said you should pick classes that start early.

- a) [CO1] Who's method will you **select** in order to complete the classes of maximum faculties? 01
- b) [CO1] Using the method you mentioned in (a) **simulate** and find out the maximum number of classes you can do from the following schedule and also find out the initials of the faculties. 06

| Faculty | Start Time | End Time |
|---------|------------|----------|
| MIBA    | 2          | 11       |
| MZU     | 3          | 4        |
| MNR     | 7          | 9        |
| AGD     | 11         | 12       |
| FGZ     | 6          | 8        |
| MNR     | 4          | 7        |
| SBD     | 8          | 11       |

- c) [CO1] As one student cannot cover all the classes so you want to determine the minimum number of students needed so that all the classes are covered. **Explain** your algorithm in a pseudocode/flow-chart/ step-by-step instructions format. 03

3. Karen started a website that displays different advertisements. She wants to maximize her advertisement revenue. Each month she receives a list of advertisements, each with a certain value (revenue) and a size (space taken on the webpage). The webpage has a limited amount of space, and she selects a subset of advertisements to display in order to maximize her total revenue, while staying within the available space. No advertisement is selected more than once, and each advertisement that is selected has to be displayed in full.

For example, following is a list of advertisements she received for the month of August:

|                                                                                                                                                                         |                                                                                                                   |
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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|

a) [CO1] Suppose, you are trying to come up with a Dynamic Programming approach to solve this problem. You build a table, each cell representing the solution (optimal revenue value) to a subproblem. **Give** the formula for filling up each entry of the table. **02**

b) [CO1] Suppose in the month of September, she received the following list of advertisements: **06**  
 Advertisements:  
 Ad 1 - Value: \$1, Size: 1  
 Ad 2 - Value: \$4, Size: 3  
 Ad 3 - Value: \$5, Size: 4  
 Ad 4 - Value: \$7, Size: 5  
 The maximum available space is 7.

**Fill** the following table according to the formula you developed in question (a). Then write the set of advertisements Karen should select.

| Value | Size | Item | Maximum Available Space |   |   |   |   |   |   |   |
|-------|------|------|-------------------------|---|---|---|---|---|---|---|
|       |      |      | 0                       | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 0     | 0    | 0    |                         |   |   |   |   |   |   |   |
| 1     | 1    | 1    |                         |   |   |   |   |   |   |   |
| 4     | 3    | 2    | 0                       | 1 | 1 | 4 | 5 | 5 | 5 | 5 |
| 5     | 4    | 3    |                         |   |   |   |   |   |   |   |
| 7     | 5    | 4    |                         |   |   |   |   |   |   |   |

c) [CO3] Suppose, Karen discovered that the top half of some of the advertisements contain nothing. So, from the month of October she is implementing a new feature, the size of at most K items can be changed to half of its original size. Revenue will stay the same. For example, **02**  
 There are four advertisements to select from,  
 Value = [17, 20, 10, 15]  
 Required Space = [4, 2, 7, 5]  
 Max Available Space = 4  
 K = 1

**Output: 37**

Explanation: Without changing the required space of any item, the solution would be taking only Ad-2, giving a revenue of 20. But if we change the size of Ad-1 into half of its original size, we can take both Ad-1 and 2. Then the total revenue is 37, which is the maximum.

Think about how you would solve this problem using a Dynamic Programming approach. Then, **write** a recursive formulation of your solution or **explain** your idea in pseudocode/flow-chart/step-by-step instructions format.

4. The white council once decided to install bidirectional rail tracks to connect several realms of middle earth. Upon their request, the elves and the dwarves devised a plan. The following table lists the tracks that can be constructed.

| From     | To       | Length of the track | Construction cost | Construction time |
|----------|----------|---------------------|-------------------|-------------------|
| Arnor    | Fangorn  | 25                  | 70                | 80                |
| Arnor    | Lindon   | 20                  | 40                | 30                |
| Arnor    | Mirkwood | 30                  | 80                | 70                |
| Fangorn  | Gondor   | 10                  | 30                | 40                |
| Fangorn  | Lindon   | 40                  | 90                | 110               |
| Fangorn  | Mirkwood | 20                  | 60                | 50                |
| Gondor   | Harad    | 25                  | 100               | 90                |
| Gondor   | Lindon   | 45                  | 110               | 100               |
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| Gondor   | Rhun     | 40                  | 120               | 130               |
| Harad    | Mirkwood | 45                  | 150               | 180               |
| Mirkwood | Rhun     | 25                  | 180               | 150               |

As there is only one engine available for the task, simultaneous construction of multiple tracks is not possible. Going through the plan, the council decided that the following tracks **must be constructed**.

- Mirkwood to Rhun
- Fangorn to Lindon

As the election is nearby, the council wants to connect all the realms with **minimum total construction time**. Now answer the following questions.

- [CO1] Which other tracks should they select for construction? **Show** a simulation of your procedure. **05**
- [CO1] **Write** the total construction time of the selected tracks. No two tracks are constructed simultaneously. **01**
- [CO1] **Calculate** the total construction cost required. **01**
- [CO1] After these tracks have been constructed, what would be the minimum distance of Harad from all other realms? Just **write** the distance from each realm. **03**

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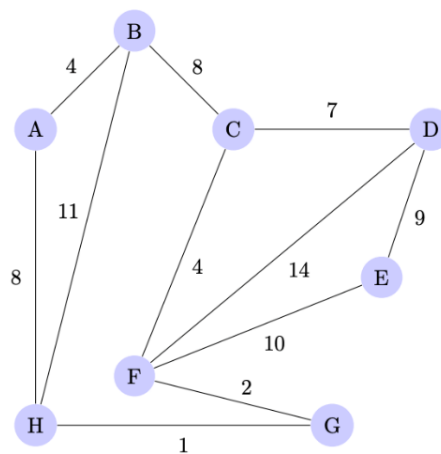
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Full Marks: 40

**CSE 221: Algorithms**

Answer the following questions.  
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|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1



*Figure 1: A Transportation Network*

- |            |                                                                                                                                                                                                                                                                        |           |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>CO2</b> | a. For the given transportation network, find a configuration connecting all regions with minimal cost. <b>Demonstrate</b> the steps and the algorithm you used to select connections.                                                                                 | <b>05</b> |
| <b>CO1</b> | b. Based on the result from part (a), <b>calculate</b> the total cost of your configuration and explain in short, why it represents the minimum cost required to connect all regions.                                                                                  | <b>02</b> |
| <b>CO3</b> | c. Now, imagine the road between Region C and Region F became <b>unavailable</b> due to maintenance. How would this impact overall connectivity? Propose an alternative solution to maintain network connectivity with minimal disruption, and explain your reasoning. | <b>03</b> |

- 2 You and your friend learned about single-source shortest path algorithms and are now given a directed weighted graph (*figure 2.*) to implement your algorithm. However, **the edge (2,4)** has an undefined weight (**Weight: X**).

*Please continue to the next page.*

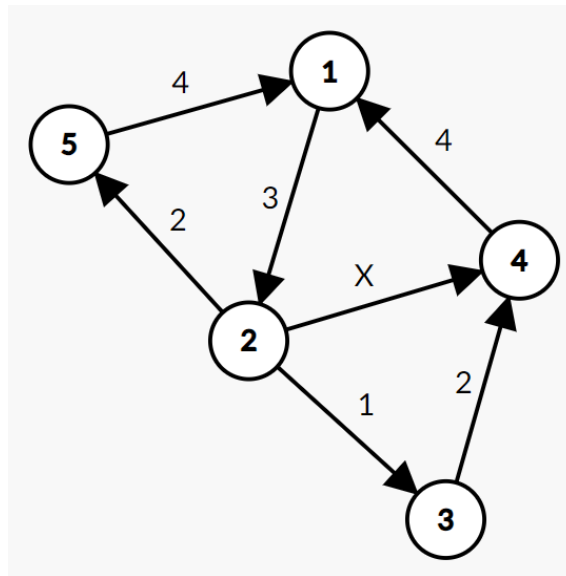


Figure 2: A Directed Weighted Graph for Question 2

- a. If the edge weight  $X = 1$ , then simulate the Dijkstra Algorithm with the necessary calculation needed to find the shortest path from **Source Node 1** to every node in the graph. Mention the minimum distances you could find. **03**
- CO2**
- b. Now, with the weight of edge (2,4) set to  $X = -9$ , prove or disprove the statement: 'This graph's single-source shortest path is unsolvable from source 1, and only Bellman-Ford can identify this issue.' Simulate/Give logical reasoning to support your answer. **04**
- CO1**
- c. If all roads have the same travel time, can an algorithm with time complexity  $O(V+E)$  be used to find the best route? Explain your idea in brief statements/pseudocode/flow-chart/step-by-step instructions. **03**
- CO1**
- 3 You are given two strings, 'cbabdc**b**' and 'badc**b**c', and tasked with finding a common sequence with the longest length. Initially, you considered a recursive solution, but person 'X' suggested using a dynamic programming approach instead. **05**
- CO3**
- a. Find the longest common sequence using the approach proposed by person 'X'. **Show** the necessary **simulation** to find the answer. **05**
- CO3**
- b. Additionally, you want to send these strings as an encoded message. After merging them into one string, 'cbabdc**b**badc**b**c', now you need an efficient text-encoding algorithm with minimal space requirement. **05**
- Simulate the encoding process,
  - construct any needed tables or trees,
  - generate the codeword for each character,
  - show the encoded message and
  - the total number of bits required to store it.

4

You are now attending the CSE221 final exam and there is a limited time to solve the questions. Each question requires a different amount of time and offers varying marks. After answering some questions, you suddenly realized that only 6 minutes remained. You aim to maximize the total marks you can score, given the constraints on time.

Questions and their properties:

- Question 4: Time = 4 minutes, Marks = 7
- Question 5: Time = 2 minutes, Marks = 9
- Question 6: Time = 3 minutes, Marks = 10
- Question 7: Time = 5 minutes, Marks = 8
- Question 8: Time = 6 minutes, Marks = 12
- Question 9: Time = 1 minute, Marks = 3

a. If your faculty doesn't give partial marks for incomplete answers, which combination of questions maximizes your total marks without exceeding the time limit? Hurry up, you need a quick, efficient approach—no time for brute force. **Show** a full detailed calculation. **04**

CO3

b. Fortunately, your faculty enters the hall and announces a new rule: partial marks will be given for partially completed answers. If you can solve part of a question, you'll get marks proportional to the time you spend on it. For example, solving half of a 4-minute question gives you half the marks. **04**

CO3

Now, with this new opportunity, how does your strategy shift? How will you maximize your score within the time limit, factoring in that you can now earn something even if you don't fully complete a question? **Calculate** the maximum marks you can score with this fortunate announcement. Did you score more than the previous scenario?

c. How does Dynamic Programming have a lower time complexity than brute force or recursive solutions? Explain briefly. **02**

CO2





**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Final Exam  
 Duration: 1 Hour 50 Minutes

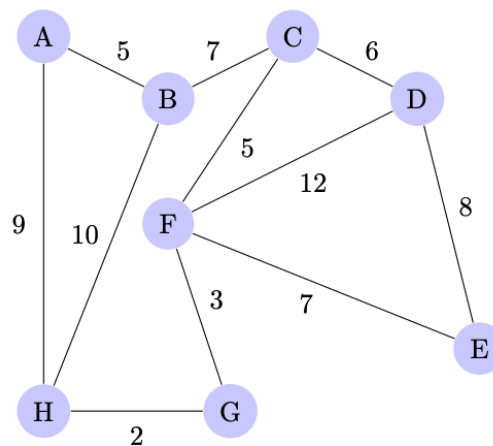
Semester: Summer 2024  
 Full Marks: 40

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1



*Figure 3: A Transportation Network*

- |            |                                                                                                                                                                                                                                                                        |           |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>CO2</b> | a. For the given transportation network, find a configuration connecting all regions with minimal cost. <b>Demonstrate</b> the steps and the algorithm you used to select connections.                                                                                 | <b>05</b> |
| <b>CO1</b> | b. Based on the result from part (a), <b>calculate</b> the total cost of your configuration and explain in short, why it represents the minimum cost required to connect all regions.                                                                                  | <b>02</b> |
| <b>CO3</b> | c. Now, imagine the road between Region C and Region D became <b>unavailable</b> due to maintenance. How would this impact overall connectivity? Propose an alternative solution to maintain network connectivity with minimal disruption, and explain your reasoning. | <b>03</b> |

- 2 You and your friend learned about single-source shortest path algorithms and are now given a directed weighted graph (*figure 2.*) to implement your algorithm. However, **the edge (2,5)** has an undefined weight (**Weight: X**).

*Please continue to the next page*

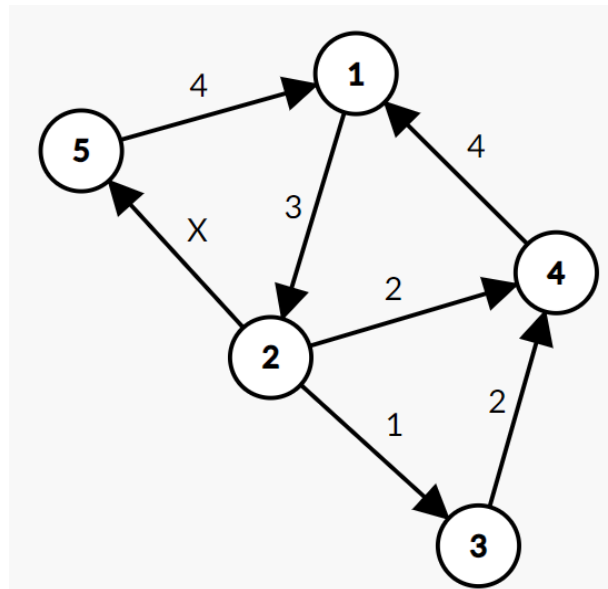


Figure 4: A Directed Weighted Graph for Question 2

- a. If the edge weight  $X = 2$ , then simulate the Dijkstra Algorithm with the necessary calculation needed to find the shortest path from **Source Node 1** to every node in the graph. Mention the minimum distances you could find. **03**
- CO2**
- b. Now, with the weight of edge (2,5) set to  $X = -11$ , prove or disprove the statement: 'This graph's single-source shortest path is unsolvable from source 1, and only Bellman-Ford can identify this issue.' Simulate/Give logical reasoning to support your answer. **04**
- CO1**
- c. If all roads have the same travel time, can an algorithm with time complexity  $O(V+E)$  be used to find the best route? Explain your idea in brief statements/pseudocode/flow-chart/step-by-step instructions. **03**
- CO1**
- 3 You are given two strings, 'abcdefg' and 'bfegba', and tasked with finding a common sequence with the longest length. Initially, you considered a recursive solution, but person 'X' suggested using a dynamic programming approach instead.
- CO3** a. Find the longest common sequence using the approach proposed by person 'X'. **Show** the necessary **simulation** to find the answer. **05**
- CO3** b. Additionally, you want to send these strings as an encoded message. After merging them into one string, 'abcdefgbfegba', now you need an efficient text-encoding algorithm with minimal space requirement. **05**
- Simulate the encoding process,
  - construct any needed tables or trees,
  - generate the codeword for each character,
  - show the encoded message and
  - the total number of bits required to store it.

- 4 You are now attending the CSE221 final exam and there is a limited time to solve the questions. Each question requires a different amount of time and offers varying marks. After answering some questions, you suddenly realized that only 6 minutes remained. You aim to maximize the total marks you can score, given the constraints on time.

Questions and their properties:

- Question 4: Time = 5 minutes, Marks = 8
- Question 5: Time = 3 minutes, Marks = 7
- Question 6: Time = 4 minutes, Marks = 12
- Question 7: Time = 6 minutes, Marks = 10
- Question 8: Time = 2 minutes, Marks = 6
- Question 9: Time = 1 minute, Marks = 5

- a. If your faculty doesn't give partial marks for incomplete answers, which combination of questions maximizes your total marks without exceeding the time limit? Hurry up, you need a quick, efficient approach—no time for brute force. **Show** a full detailed calculation. **05**
- CO3**

- b. Fortunately, your faculty enters the hall and announces a new rule: partial marks will be given for partially completed answers. If you can solve part of a question, you'll get marks proportional to the time you spend on it. For example, solving half of a 4-minute question gives you half the marks. **05**
- CO3**

Now, with this new opportunity, how does your strategy shift? How will you maximize your score within the time limit, factoring in that you can now earn something even if you don't fully complete a question? **Calculate** the maximum marks you can score with this fortunate announcement. Did you score more than the previous scenario?

- c. How does Dynamic Programming have a lower time complexity than brute force or recursive solutions? Explain briefly. **02**
- CO2**



**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Semester Final Exam  
 Duration: 2 Hours

Semester: Summer 2025  
 Full Marks: 45

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a.** A directed graph has the following vertices:  
 [CO1] 1, 2, 3, 4, 5, 6, 7, 8

Following is the adjacency list for the graph:

1: [2, 3]  
 2: [4]  
 3: [1, 5]  
 4: [6]  
 5: [6]  
 6: [7]  
 7: [5, 8]  
 8: [ ]

- i. Using the above information, **draw** the directed graph. **02**  
 ii. **Identify** and write all the Strongly Connected Components containing at least 2 nodes. **02**  
 [No need to show any process.]

- b.** Consider the following everyday morning activities:

- [CO1] → Waking Up  
 → Brushing  
 → Leaving the house  
 → Taking a shower  
 → Getting dressed  
 → Having breakfast

Constraints:

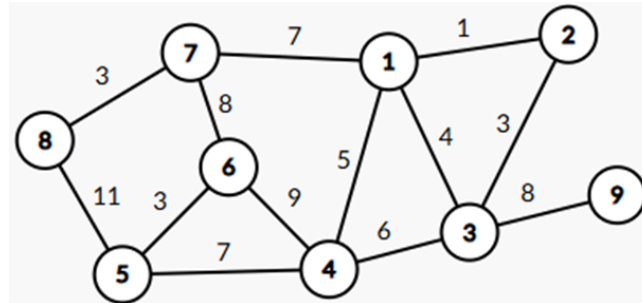
- You must not get dressed before taking a shower
- You are free to have breakfast and then brush, or the opposite
- After waking up: you can either take a shower and then brush, or the opposite
- You must not leave the house without getting dressed
- Waking up is the first thing you do, and leaving the house is the last activity

- i. **Draw** a directed graph representing the activities as nodes, and constraints as edges. **03**  
 ii. From the above activities, **write** two different topologically sorted sequences of activities. **02**  
 iii. **Explain** in short, how you would use DFS to topologically sort the activities. **01**

2

The city is struck by a massive flood. Several shelters have been established. Rescue teams are coordinating supplies, rescue boats etc. To ensure communication, the government wants to set up temporary bridges between shelters so that all shelters remain connected.

For example, in the graph below, nodes represent shelters. The cost of setting up bridges between two shelters are given as weighted edges.



a. **Simulate** a suitable algorithm to find which bridges to build so that all shelters remain connected at the minimum possible cost. Also, report the total cost of this arrangement. 04

[CO1]

b. However, new information arrived that shelter 3 is already equipped with satellite communication devices, which make direct connections to this shelter cheaper than normal. Propose a modification to your algorithm in Q2(a) to directly connect (by building bridges) as many shelters as possible with shelter 3, even if they are not the lowest-weight edges in the overall graph. Just **explain** the modification. 03

[CO3]

c. After connecting all the shelters, you have decided to make shelter 2 as the base of operations. So you want to connect all shelters with shelter 2 at minimum possible cost. Do the same set of bridges you have built for Q2(a) also serve this purpose? **Explain** in short with proper reasoning. 03

[CO2]

3

a. During the July festival, organizers want to arrange a drone show. There will be  $N$  drones carrying lights over an  $N \times N$  aerial stage. The drones must form a symmetrical light display, but no two drones can align vertically or horizontally. 02

[CO1]

If the number of drones is 6, **draw** a valid arrangement of drones in a 6x6 grid.

02

b. A delivery robot works in a storage area organized in rows and columns of delivery packages, where each package can be thought of as a cell. There is only one free spot (cell) that allows the robot to slide packages around. Its goal is to arrange all the packages in a specific order so that they are ready for shipping.

[CO1]

While trying different moves, the robot keeps going back to layouts it has already tried, getting stuck in a loop and making no progress.

1. How can the robot avoid repeating the same arrangements again and again? **Explain** within two sentences. 02

2. **Name** an algorithm that will help to find the shortest sequence of moves to reach the correct order of packages. 01

**4**

You are designing a data compression and transmission system for a sensor network that monitors environmental conditions. Each type of sensor reading has a frequency of occurrence and an impact factor indicating its importance for critical decision-making.

Your goals are to compress the data efficiently for transmission using a variable-length encoding system and maximize the total impact factor transmitted within the system's data capacity.

Sensor Data Table:

| Sensor Reading | Frequency of occurrence | Impact Factor |
|----------------|-------------------------|---------------|
| Temperature    | 45                      | 10            |
| Humidity       | 13                      | 6             |
| Pressure       | 12                      | 8             |
| Light          | 16                      | 7             |
| CO2            | 9                       | 5             |
| Wind Speed     | 5                       | 4             |
| Rainfall       | 7                       | 6             |
| UV Index       | 3                       | 3             |

a. **Construct** a Huffman tree for these sensor readings based on their frequencies of occurrence. Then **assign** binary codes to each reading. **05**

b. **Calculate** the total number of bits required for the transmission of all readings. **02**

c. Suppose, the system can transmit only 15 units of data. Each sensor reading generates 1 (one) unit of data per occurrence. You can partially transmit the readings but its impact factor drops proportionally if you do so. **03**

**Determine** which readings to transmit and with what frequency, to maximize the total impact factor within the 15-unit data limit. **Calculate** the maximum total impact factor.

**5**

A genetics research team is studying DNA sequences to identify whether mutations have occurred during sequencing. When an experimental DNA reading is obtained, it is compared against a known reference gene to evaluate their similarity. However, sequencing errors or natural mutations may lead to insertions, deletions, or substitutions of nucleotides.

To measure this similarity, two approaches are commonly used. The Longest Common Subsequence (LCS) measures similarity by finding the longest subsequence that appears in both sequences without rearranging their order. On the other hand, the Edit Distance measures dissimilarity by calculating the minimum number of edits (insertions, deletions,

or substitutions) required to transform one sequence into the other. Both can be solved using Dynamic Programming as there are many overlapping subproblems.

You are provided with the following two DNA sequences:

Experimental reading (Read): **ACGTA**

Reference gene (Ref): **AGCTG**

Suppose,  $m = \text{length}(\text{Read})$  and  $n = \text{length}(\text{Ref})$ .

For both LCS, and Edit Distance

- The dimension of the dp table is  $(m+1)*(n+1)$
- The final answer is stored in cell  $\text{dp}[m][n]$

**a.** Recurrence relation for Longest Common Subsequence:  
[CO1]

$$\text{dp}[i][j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ 1 + \text{dp}[i-1][j-1] & \text{if } \text{Read}[i-1] = \text{Ref}[j-1] \\ \max(\text{dp}[i-1][j], \text{dp}[i][j-1]) & \text{otherwise} \end{cases}$$

Here, each cell  $\text{dp}[i][j]$  stores the length of the LCS between the first  $i$  characters of Read and the first  $j$  characters of Ref.

- (i) **Construct** the dp table to find the LCS of the provided DNA sequences (Read and Ref) by following this recurrence relation. **04**
- (ii) **Determine** the length of LCS from the dp table and **write** a valid LCS. **01**

**b.** Recurrence relation for Edit Distance:  
[CO3]

$$\text{dp}[i][j] = \begin{cases} i & \text{if } j = 0 \\ j & \text{if } i = 0 \\ \text{dp}[i-1][j-1] & \text{if } \text{Read}[i-1] = \text{Ref}[j-1] \\ 1 + \min(\text{dp}[i-1][j], \text{dp}[i][j-1], \text{dp}[i-1][j-1]) & \text{otherwise} \end{cases}$$

Here, each cell  $\text{dp}[i][j]$  stores the minimum number of edits required to transform the first  $i$  characters of Read into the first  $j$  characters of Ref.

- (i) **Construct** the dp table to find the Edit Distance between the provided DNA sequences (Read and Ref) by following this recurrence relation. **03**
- (ii) From the dp table, **determine** the minimum number of edits required to transform one sequence into the other. **01**

**c.** Prove the following using the answers from Q5(a-ii) and Q5(b-ii).  
[CO2] **EditDistance(Read, Ref)  $\leq m + n - 2 \times \text{LCS}(\text{Read, Ref})$**



**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Semester Final Exam  
 Duration: 2 Hours

Semester: Summer 2025  
 Full Marks: 45

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a.** A directed graph has the following vertices:  
 [CO1] 1, 2, 3, 4, 5, 6, 7, 8

Following is the adjacency list for the graph:

1: [2]  
 2: [3]  
 3: [1, 4]  
 4: [5]  
 5: [6]  
 6: [4]  
 7: [6, 8]  
 8: [7]

- i. Using the above information, **draw** the directed graph. **02**  
 ii. **Identify** and write all the Strongly Connected Components containing at least 2 nodes. **02**  
 [No need to show any process.]

- b.** Consider the following nighttime activities:  
 [CO1]

- Having dinner
- Brushing teeth
- Going to bed
- Turning off the lights
- Changing into pajamas
- Fending off the ghosts

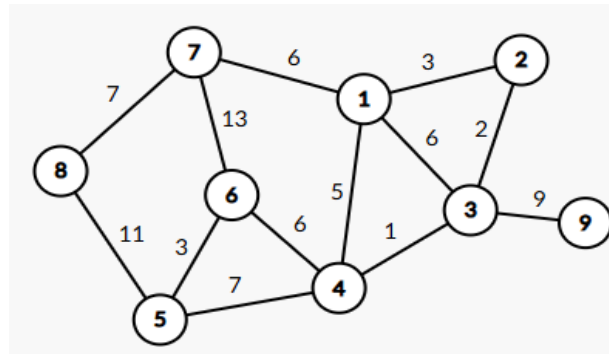
Constraints:

- You must not go to bed before brushing your teeth
- You must change into pajamas before going to bed
- You are free to have dinner before or after fending off the ghosts
- You must turn off the lights before going to bed
- Fending off the ghosts must be done before turning off the lights
- Going to bed is the last activity

- i. **Draw** a directed graph representing the activities as nodes, and constraints as edges. **03**  
 ii. From the above activities, **write** two different topologically sorted sequences of activities. **02**  
 iii. **Explain** in short, how you would use DFS to topologically sort the activities. **01**

- 2** The city is struck by a massive flood. Several shelters have been established. Rescue teams are coordinating supplies, rescue boats etc. To ensure communication, the government wants to set up temporary bridges between shelters so that all shelters remain connected.

For example, in the graph below, nodes represent shelters. The cost of setting up bridges between two shelters are given as weighted edges.



- a.** **Simulate** a suitable algorithm to find which bridges to build so that all shelters remain connected at the minimum possible cost. Also, report the total cost of this arrangement. **04**  
[CO1]
- b.** However, new information arrived that shelter 2 is already equipped with satellite communication devices, which make direct connections to this shelter cheaper than normal. Propose a modification to your algorithm in Q2(a) to directly connect (by building bridges) as many shelters as possible with shelter 2, even if they are not the lowest-weight edges in the overall graph. Just **explain** the modification. **03**  
[CO3]
- c.** After connecting all the shelters, you have decided to make shelter 4 as the base of operations. So you want to connect all shelters with shelter 4 at minimum possible cost. Do the same set of bridges you have built for Q2(a) also serve this purpose? **03**  
[CO2]  
**Explain** in short with proper reasoning.
- 3 a.** During the July festival, organizers want to arrange a drone show. There will be  $N$  drones carrying lights over an  $N \times N$  aerial stage. The drones must form a symmetrical light display, but no two drones can align vertically or horizontally. **02**  
[CO1]  
If the number of drones is 6, **draw** a valid arrangement of drones in a 6x6 grid.
- b.** A delivery robot works in a storage area organized in rows and columns of delivery packages, where each package can be thought of as a cell. There is only one free spot (cell) that allows the robot to slide packages around. Its goal is to arrange all the packages in a specific order so that they are ready for shipping. **01**  
[CO1]  
While trying different moves, the robot keeps going back to layouts it has already tried, getting stuck in a loop and making no progress.
1. How can the robot avoid repeating the same arrangements again and again? **02**  
**Explain** within two sentences.
  2. **Name** an algorithm that will help to find the shortest sequence of moves to reach the correct order of packages. **01**

- 4** You are designing a data compression and transmission system for a sensor network that monitors environmental conditions. Each type of sensor reading has a frequency of occurrence and an impact factor indicating its importance for critical decision-making.

Your goals are to compress the data efficiently for transmission using a variable-length encoding system and maximize the total impact factor transmitted within the system's data capacity.

Sensor Data Table:

| Sensor Reading | Frequency of occurrence | Impact Factor |
|----------------|-------------------------|---------------|
| Temperature    | 50                      | 12            |
| Humidity       | 10                      | 7             |
| Pressure       | 15                      | 9             |
| Light          | 20                      | 8             |
| CO2            | 8                       | 6             |
| Wind Speed     | 6                       | 5             |
| Rainfall       | 12                      | 7             |
| UV Index       | 4                       | 3             |

- a. **Construct** a Huffman tree for these sensor readings based on their frequencies of occurrence. Then **assign** binary codes to each reading. **05**  
[CO1]
- b. **Calculate** the total number of bits required for the transmission of all readings. **02**  
[CO1]
- c. Suppose, the system can transmit only 20 units of data. Each sensor reading generates 1 (one) unit of data per occurrence. You can partially transmit the readings but its impact factor drops proportionally if you do so. **03**  
[CO1]
- Determine** which readings to transmit and with what frequency, to maximize the total impact factor within the 20-unit data limit. **Calculate** the maximum total impact factor.

- 5** A genetics research team is studying DNA sequences to identify whether mutations have occurred during sequencing. When an experimental DNA reading is obtained, it is compared against a known reference gene to evaluate their similarity. However, sequencing errors or natural mutations may lead to insertions, deletions, or substitutions of nucleotides.

To measure this similarity, two approaches are commonly used. The Longest Common Subsequence (LCS) measures similarity by finding the longest subsequence that appears in both sequences without rearranging their order. On the other hand, the Edit Distance

measures dissimilarity by calculating the minimum number of edits (insertions, deletions, or substitutions) required to transform one sequence into the other. Both can be solved using Dynamic Programming as there are many overlapping subproblems.

You are provided with the following two DNA sequences:

Experimental reading (Read): **ACGTA**

Reference gene (Ref): **GACGC**

Suppose,  $m = \text{length}(\text{Read})$  and  $n = \text{length}(\text{Ref})$ .

For both LCS, and Edit Distance

- The dimension of the dp table is  $(m+1)*(n+1)$
- The final answer is stored in cell  $\text{dp}[m][n]$

**a.** Recurrence relation for Longest Common Subsequence:  
[CO1]

$$dp[i][j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ 1 + dp[i-1][j-1] & \text{if } \text{Read}[i-1] = \text{Ref}[j-1] \\ \max(dp[i-1][j], dp[i][j-1]) & \text{otherwise} \end{cases}$$

Here, each cell  $\text{dp}[i][j]$  stores the length of the LCS between the first  $i$  characters of Read and the first  $j$  characters of Ref.

(i) **Construct** the dp table to find the LCS of the provided DNA sequences (Read and Ref) by following this recurrence relation. **04**

(ii) **Determine** the length of LCS from the dp table and **write** a valid LCS. **01**

**b.** Recurrence relation for Edit Distance:  
[CO3]

$$dp[i][j] = \begin{cases} i & \text{if } j = 0 \\ j & \text{if } i = 0 \\ dp[i-1][j-1] & \text{if } \text{Read}[i-1] = \text{Ref}[j-1] \\ 1 + \min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) & \text{otherwise} \end{cases}$$

Here, each cell  $\text{dp}[i][j]$  stores the minimum number of edits required to transform the first  $i$  characters of Read into the first  $j$  characters of Ref.

(i) **Construct** the dp table to find the Edit Distance between the provided DNA sequences (Read and Ref) by following this recurrence relation. **03**

(ii) From the dp table, **determine** the minimum number of edits required to transform one sequence into the other. **01**

**c.** Prove the following using the answers from Q5(a-ii) and Q5(b-ii). **01**  
[CO2] **EditDistance(Read, Ref)  $\leq m + n - 2 \times \text{LCS}(\text{Read, Ref})$**